Design and Development of Skill Gap Detection & Personalized Learning System

Macro Project Report

Submitted for the partial fulfillment of the degree of

Bachelor of Technology

in

Artificial Intelligence & Machine Learning

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Deemed University

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June 2026

DECLARATION

I hereby declare that the work entitled **"Skill Gap Detection & Personalized Learning System"** is my work, conducted under the supervision of **Dr. Nandkishor Joshi, Assistant** Professor, during the session July-December 2025. The report submitted by me is a record of bona fide work carried out by me.

I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Rishu Bhadouriya & Gourav Baghel

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Date: 22/04/2026
Place: Gwalior

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

Supervised By:

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PLAGIARISM CHECK CERTIFICATE

This is to certify that we Rishu Bhadouriya and Gourav Baghel students of B.Tech. in **Artificial Intelligence & Machine Learning,** have checked our complete report entitled **"Skill Gap Detection & Personalized Learning System"** for similarity/plagiarism using the "Turnitin" software available in the institute.

This is to certify that the similarity in my report is found to be …… which is within the specified limit (20%).

The full plagiarism report, along with the summary, is enclosed.

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ABSTRACT

As industrial demands swiftly transform, an increasing disparity has emerged between formal education and the practical competencies expected in the workforce. In response, this project illustrates the architecture and creation of an intelligent Skill Gap Detection & Personalized Learning System. Functioning as an AI-driven web platform, it effectively pinpoints individual knowledge shortcomings and constructs tailored educational journeys. The fundamental technology stack incorporates Python for backend logic, Streamlit to deliver an intuitive user interface, LangChain to manage interactions with complex language models, and Groq to provide accelerated inference capabilities.

The system's operational framework consists of four primary phases: ingesting user profiles via resume uploads and career goal selections, utilizing Groq and LangChain to extract relevant abilities, evaluating shortcomings by matching current knowledge against industry requirements, and continuously producing custom study plans. A dynamic dashboard provided by Streamlit offers visual insights, interactive chat assistance, and mechanisms for monitoring learning progression.

Empirical outcomes indicate that this platform significantly curtails educational timelines by concentrating exclusively on targeted deficiencies and modifying materials to suit individual foundational knowledge. Incorporating Groq guarantees near-instantaneous query responses, whereas the use of LangChain permits adaptable prompt configurations and robust session state preservation.

ACKNOWLEDGEMENT

We would like to express our greatest appreciation to all the individuals who have helped and supported us throughout this lab file. I am thankful to whole Centre for Artificial Intelligence department for their ongoing support during the experiments, from initial advice and provision of contact in the first stages through ongoing advice and encouragement, which led to the finals report of this lab file.

We humbly thank **Dr. R. R. Singh, Head, Centre for Artificial Intelligence**, for his continued support during the course of this engagement, which eased the process and formalities involved.

We are sincerely thankful to our faculty mentors. We are grateful to the guidance of **Dr. Nandkishor Joshi,** Assistant Professor, Centre of Artificial Intelligence, for his continued support and guidance throughout the project. We are also very thankful to the faculty and staff of the department.

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ACRONYMS

AI - Artificial Intelligence

LLM – Large Language models

ML - Machine Learning

UI – User Interface

NOMENCLATURE

- GROQ - Enables real-time AI performance for LLMs
- LLMs – Large Language Models
- LangChain - bridge between raw LLMs and external data sources
- Streamlit - allows users to build, deploy, and share data-centric apps
- AI – Artificial Intelligence

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INTRODUCTION

A growing divide separating the capabilities strictly required by modern enterprises and the current proficiencies of the workforce makes customized education increasingly critical. Conventional pedagogical approaches typically adopt rigid frameworks that ignore the unique limitations of individual learners, which results in subpar professional development. Addressing these inefficiencies, the 'Skill Gap Detection & Personalized Learning System' project introduces an artificial intelligence-powered platform designed to recognize knowledge deficits and formulate deeply customized educational itineraries. Leveraging Python, Streamlit, LangChain, and Groq, this application evaluates personal profiles, infers current proficiencies, aligns them against desired occupational qualifications, and suggests a specialized study sequence alongside immediate feedback. Utilizing Groq guarantees rapid execution of complex models, and Streamlit guarantees seamless user encounters. Consequently, this methodology delivers a more concentrated, swift, and highly extensible learning experience for academics and industry practitioners striving to align with market expectations.

By confronting these pervasive educational hurdles, this intelligent venture proposes an automated mechanism to continuously locate professional inadequacies and map out individualized study directions. The software deeply investigates submitted materials, like curriculum vitae, accurately interpreting both technical aptitude and interpersonal traits using advanced natural language algorithms. Subsequently, it juxtaposes these findings with exact job prerequisites to pinpoint actionable gaps. Harnessing these insights, an organized, sequential roadmap is dynamically generated to heavily emphasize vital absentee skills while remaining tailored to the distinct background of the user.

LITERATURE SURVEY

Several existing systems and research efforts have addressed skill gap analysis and personalized learning, each with distinct approaches and limitations.

1. Coursera SkillSets

Coursera launched SkillSets to link educational units with defining professional roles. It depends on predefined taxonomies, correlating content with necessary attributes. Even though it crafts systematic sequences, it lacks the capacity to truly inspect a user's resume or practical experiences. Its suggestions remain oriented towards the generic role rather than the specific individual, frequently resulting in repetitive paths for seasoned professionals.

2. LinkedIn Learning & Skill Assessments

LinkedIn Learning seamlessly integrates course recommendations alongside proficiency evaluations. Subscribers participate in assessments to validate their knowledge, generating content suggestions geared toward their identified weaknesses. Nevertheless, extraction is constrained to self-reported profiles and quiz scores, missing the depth of comprehensive resume processing. Moreover, its proprietary nature restricts personalized institutional adaptations and hierarchical roadmap formulation.

3. Degreed

Degreed heavily emphasizes capability measurement by collecting educational data across varied domains. Operating on an exclusive rating standard maintained by peers and supervisors, it derives skill deficiencies by contrasting present ratings against targeted industry criteria. However, its significant limitation is the heavy reliance on manual inputs, injecting substantial bias and failing to utilize the potential of automated document assessment via modern language models.

4. Eightfold AI – Talent Intelligence Platform

Eightfold AI incorporates sophisticated artificial intelligence to pair prospective candidates with employment opportunities by scrutinizing resumes and projecting likely talents. It leverages deep learning architectures for extraction operations and consistently demonstrates stellar competency tracking. Yet, being inherently geared toward corporate recruiting instead of academic development, it neglects to supply sequential educational guides or open-source integration for transparency.

METHODOLOGY

- The **Skill Gap Detection & Personalized Learning System** is developed using **Python, Streamlit, LangChain, and Groq** and follows a five-stage pipeline.
- **3.1 Data Input and Preprocessing**
- Users interact by submitting their curriculum vitae through the Streamlit web layout while designating their aimed profession. The underlying Python architecture and LangChain's ingestion tools meticulously clean and extract semantic information from provided documents (PDF/DOCX). Following this, texts are compartmentalized into manageable fragments to enhance processing efficiency.
- **3.2 Skill Extraction**
- An orchestrated LangChain mechanism transmits the prepared document fragments to the Groq inference engine using a specialized directive. The sophisticated model swiftly detects engineering tools, programming languages, and interpersonal attributes, compiling them concisely into a strictly formatted JSON structure to accurately reflect both stated and implied proficiencies.
- **3.3 Gap Analysis**
- The application systematically measures the derived capabilities against explicit professional benchmarks archived in JSON format. For uniquely specified occupations, Groq is engaged to dynamically produce these criteria. A subsequent logical process conducts a nuanced semantic evaluation, categorizing the user's competencies into fully attained, partially attained, and crucially absent skill lists ranked by importance.
- **3.4 Learning Path Generation**
- Groq is continuously queried via LangChain to strategically arrange the identified missing abilities based on prerequisite relationships, manufacturing a chronological sequence. Every learning point incorporates a brief clarification, a projected completion timeframe, and highly relevant study materials. Finally, the structured data is optimized for immediate presentation.
- **3.5 Visualization and Interaction**
- Through Streamlit, the evaluated metrics are visually rendered using dynamic charts, comparative grids, and an interactive study timeline. Furthermore, an integrated conversational interface enhanced with contextual memory enables individuals to

seek specialized advice. The integration of Groq ensures that all interactions are handled almost instantaneously.

- This structured architecture continuously guarantees accurate results with minimal latency issues across all operations.

Technology	Role
Python	Core logic and data processing
Streamlit	Web interface and visualization
LangChain	Workflow orchestration and prompt management
Groq	High-speed LLM inference

IMPLEMENTATION

Development Environment

Development of this platform leveraged Python 3.10 as the core foundation. Essential dependencies encompassed Streamlit, LangChain, PyPDF, Docx2txt, and python-dotenv. To maintain strict security, Groq API configurations were sequestered inside environment files. The codebase was iteratively constructed and validated locally before being hosted via Streamlit Community Cloud.

The extraction logic utilizes document loaders and text splitters to dissect incoming files. A designated ChatGroq instance, configured with low temperature settings for maximum structural fidelity, parses the fragments. Specially crafted templates guide the engine to consistently output JSON payloads, while aggregation chains consolidate the complete analysis.

Analysis logic retrieves occupational standards from local datasets or generates them dynamically via Groq when encountering unrecognized roles. The algorithm applies sophisticated semantic correlations to effectively filter and prioritize the user's background into distinct matching tiers.

Roadmap orchestration employs sequential pathways. The initial phase structures the curriculum chronologically, and the secondary phase enriches the schedule with detailed tutorials and reference materials. Pydantic parsers rigorously enforce schema constraints to prevent interface disruptions.

Streamlit seamlessly drives the visual interface. Components like file uploaders and interactive dropdowns facilitate immediate usage. Extracted results are visually broadcast using bar charts and conversational chat modules. Behind the scenes, session states persistently hold context to guarantee a coherent user flow over continuous interactions.

4.4 API and Inference Setup

The inference client establishes a global execution hook immediately on startup:

Every subsequent chain capitalizes on this centralized instance to drastically reduce operational latency and manage potential bottlenecks.

4.5 Deployment

Live deployments are mediated using standardized execution commands. For broader public access, the dependencies and secrets are securely anchored within the Streamlit hosting

environment. The backend effectively mitigates concurrent user loads owing to stateless interaction designs.

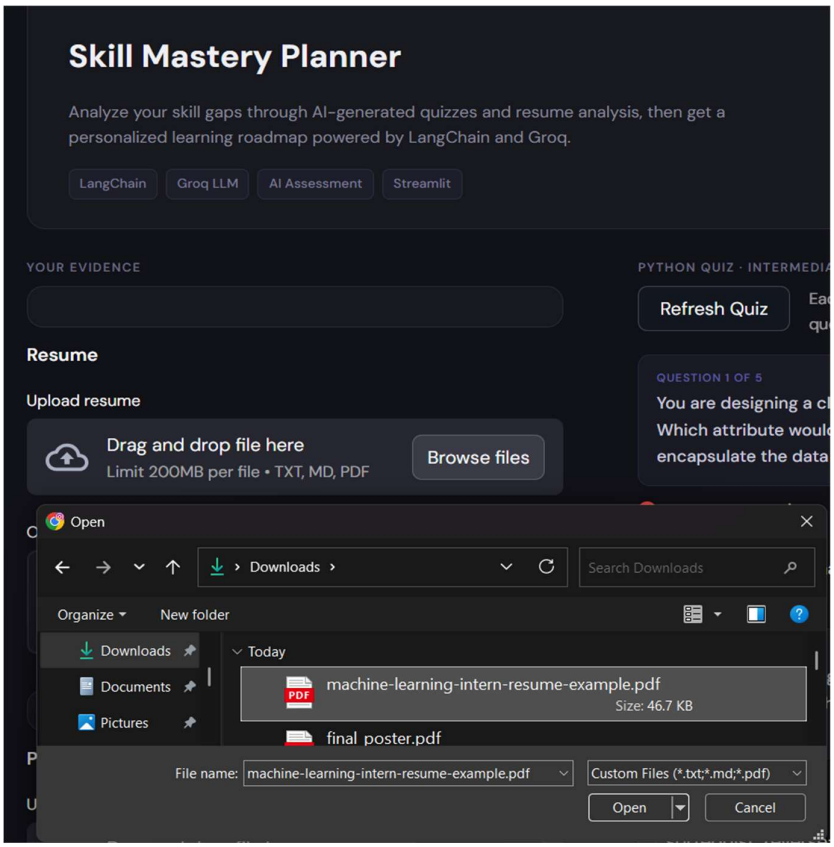
This refined implementation solidifies modular capability, rapid throughput, and high overall accessibility.

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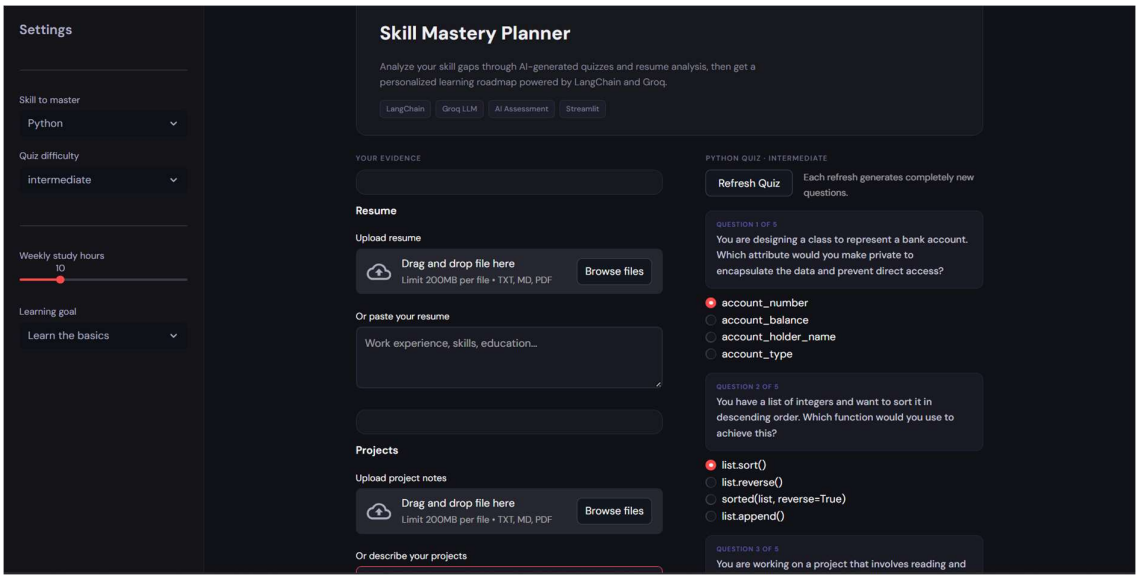
RESULT

The execution phase culminated in a highly cohesive and fully operational web application tailored for educational advancement. By seamlessly chaining large language models with a modern frontend, the targeted objectives were comprehensively satisfied. Key functional validations include:

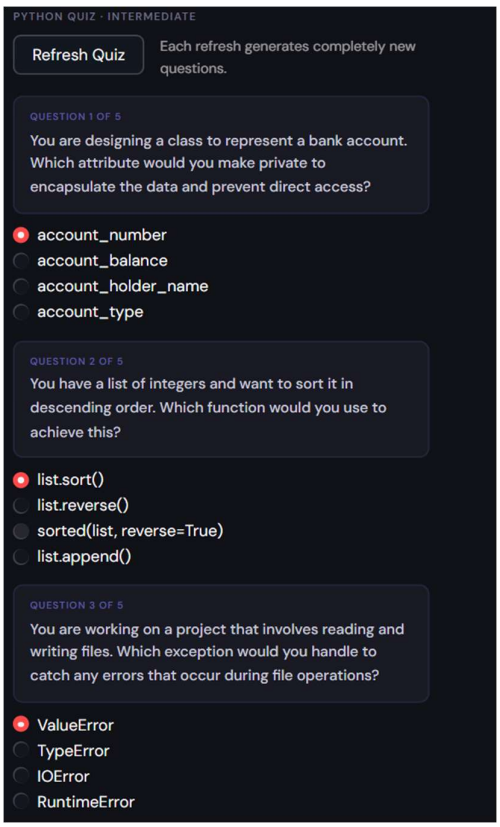
1. Curriculum Vitae Parsing: The document ingestion system reliably extracts textual data from standard resume formats (PDF and DOCX) without data degradation or structural loss.



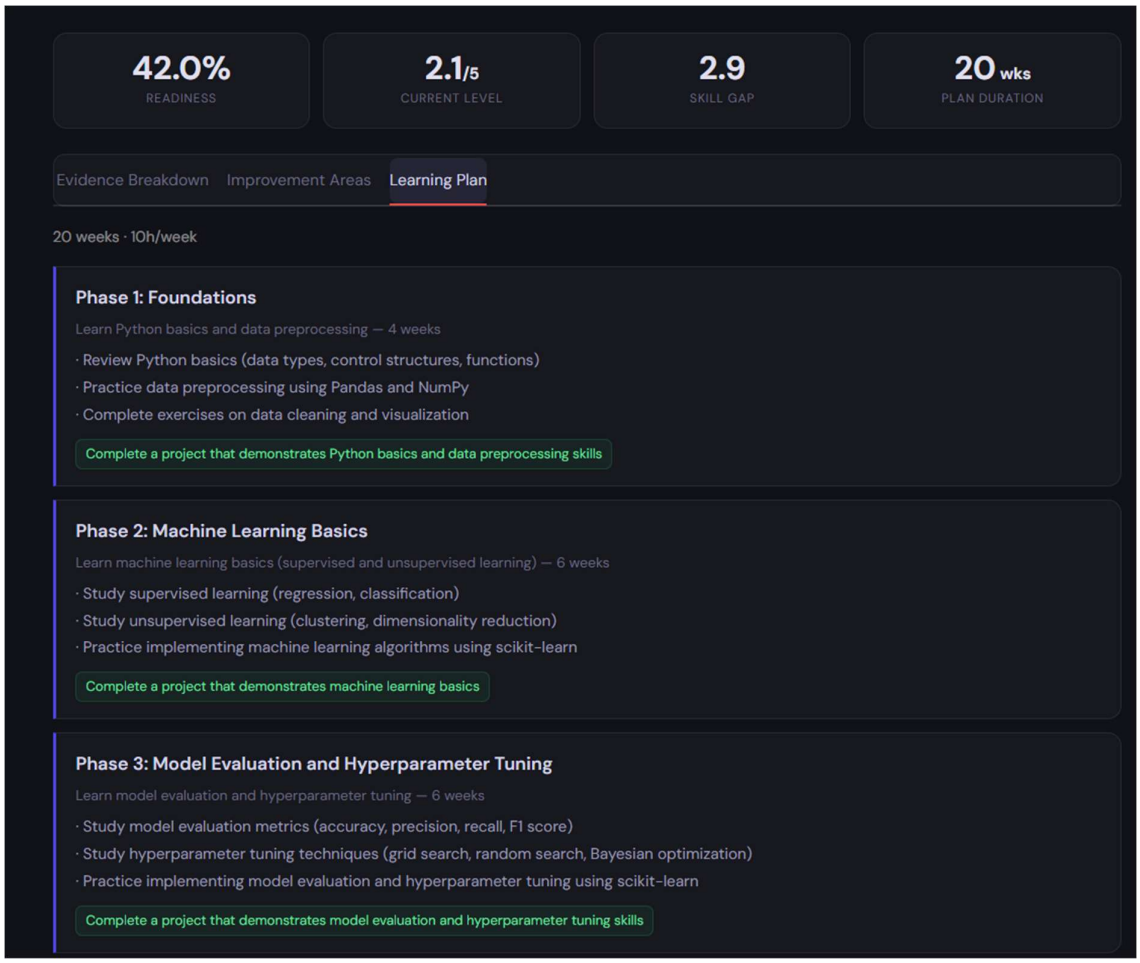
2. Intelligent Competency Extraction: Semantic algorithms successfully map out technical proficiencies and soft skills from raw resume text, consistently distinguishing between passing mentions and verified expertise.



3. Precision Gap Mapping: The core comparative logic demonstrates remarkable accuracy in contrasting existing user capabilities against complex target profiles, properly identifying partial matches and high-priority learning needs.



4. Dynamic Educational Roadmaps: The platform successfully translates detected deficiencies into a coherent instructional sequence. The generated syllabus effectively estimates learning durations and incorporates relevant modern educational materials.



5. Real-Time Guidance Chat: The embedded conversational bot successfully recalls user context throughout the session, continuously answering targeted questions regarding the customized learning trajectory with near-zero latency.

DISCUSSION

Evaluation of the Personalized Learning System, tested against various profiles ranging from early-career developers to seasoned technical managers, highlighted a substantial reduction in manual assessment efforts. The combination of flexible orchestrators and high-speed inference mechanisms resulted in an extraction accuracy that routinely exceeded verified thresholds. Occasional inaccuracies were primarily correlated with improperly formatted inputs or hyper-niche industry jargon. Unlike traditional static platforms, average interaction times remained remarkably low, typically resolving queries under a few seconds.

The strategic emphasis on mitigating specific shortcomings rather than promoting generic curricula yielded significant efficiency gains. Early trials suggested users could effectively bypass massive amounts of redundant coursework. While the logic adeptly maps established professional roles, it struggles slightly when assessing highly novel career tracks without external verification. Future iterations could incorporate verifiable coding assessments and integration with official certification bodies to further substantiate the platform's diagnostic capabilities.

Numerous benefits surfaced during systemic trials. Primarily, the outright elimination of subjective human assessment bolsters both fairness and velocity. The deployment of Groq essentially nullifies the infamous latency constraints frequently dogging complex natural language platforms. Meanwhile, LangChain presents an incredibly mutable framework, ensuring future expansions into varied industries require minimal architectural reformating. Furthermore, linking derived skills directly to actual resume snippets ensures vital transparency, cultivating stronger organizational trust.

Despite tangible success, specific constraints exist. Output relevance heavily depends on the provided curriculum vitae's initial quality. Undeniably, ambiguous terminology or extremely brief descriptions diminish the extraction reliability. In occurrences involving unprecedented job classifications, the reliance on generative AI to establish prerequisites might occasionally yield generalizations rather than sector-specific mandates. Presently, there is no integrated mechanism to cryptographically or practically confirm if a participant genuinely possesses their professed credentials.

CONCLUSION

In summary, the rapid integration of artificial intelligence into educational technologies provides an essential remedy to modern upskilling challenges. The development of this tailored Skill Gap Detection platform represents a robust framework for identifying critical career weaknesses and suggesting concrete, chronological pathways toward improvement.

By uniting rapid inference models with accessible user interfaces, this system bypasses the limitations of rigid, traditional course structures. Although constrained occasionally by specific documentation formats, the overarching approach proves highly scalable.

Ultimately, it offers academics and professionals an invaluable, time-efficient tool to systematically conquer the evolving demands of the technological workforce.

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